



Korea SKT TBC1000B

New Software Feature Test Report

Test date: 2026-06-25
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Test Summary

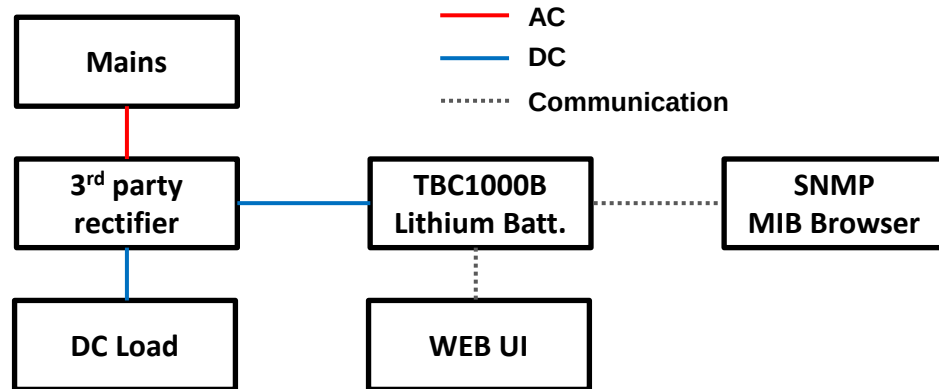


Tested Equipment	TBC1000B-NDA1 + ESM-48100C1
Tested Software Version	SMU: V100R025C10SPC210
	BMU: EN61PWRAC01136
Test Case	1. SOC 95% Charging Stop Function Test
	2. Battery Remote Disconnect Function Test
	3. Checking Battery SOC/SOH/SN via SNMP

New Software Feature Test

■ SOC 95% Charging Stop Function Test

□ Test Block diagram



□ Test Method

1. Discharge the battery to 90% SOC.
2. After discharge is complete, set the SOC charge limit to 95%.
3. Start charging.
4. Verify that charging stops at 95% SOC and the system enters Float Charge mode.

□ Test result

1. Charging stops and switches to Float Charge mode at 95% SOC, with no further charging.
2. **PASS**

New Software Feature Test

■SOC 95% Charging Stop Function Test



The screenshot shows the 'Monitoring' tab of the Enspire power system interface. The left sidebar lists various components: Site Unit, Digital Power, Lithium Battery Group, GPS, Extend LI Battery Cabinet1, UIM, and Lithium Batteries 1 through 8, along with Collection Equipment. The main panel displays 'Basic Information' for the selected component. The SOC (State of Charge) is highlighted with a red box, showing a value of 90 %.

Parameter	Value
Battery Voltage	48.8 V
Battery Current	-22.7 A
Cell Pack Voltage	49.2 V
Cell Pack Current	-23.3 A
SOC	90 %
Cell 1 Temperature	30 °C
Cell 2 Temperature	30 °C
Cell 3 Temperature	30 °C
Cell 4 Temperature	29 °C
Cell 5 Temperature	29 °C
Cell 6 Temperature	29 °C
Cell 7 Temperature	29 °C
Cell 8 Temperature	29 °C
Cell 9 Temperature	29 °C
Cell 10 Temperature	29 °C
Cell 11 Temperature	29 °C
Cell 12 Temperature	31 °C
Cell 13 Temperature	31 °C
Cell 14 Temperature	31 °C
Cell 15 Temperature	31 °C

The screenshot shows the 'Monitoring' tab of the Enspire power system interface. The left sidebar lists various components: Site Unit, Digital Power, Lithium Battery Group, GPS, Extend LI Battery Cabinet1, UIM, and Lithium Batteries 1 through 8, along with Collection Equipment. The main panel displays 'Basic Information' for the selected component. The SOC (State of Charge) is highlighted with a red box, showing a value of 90 %.

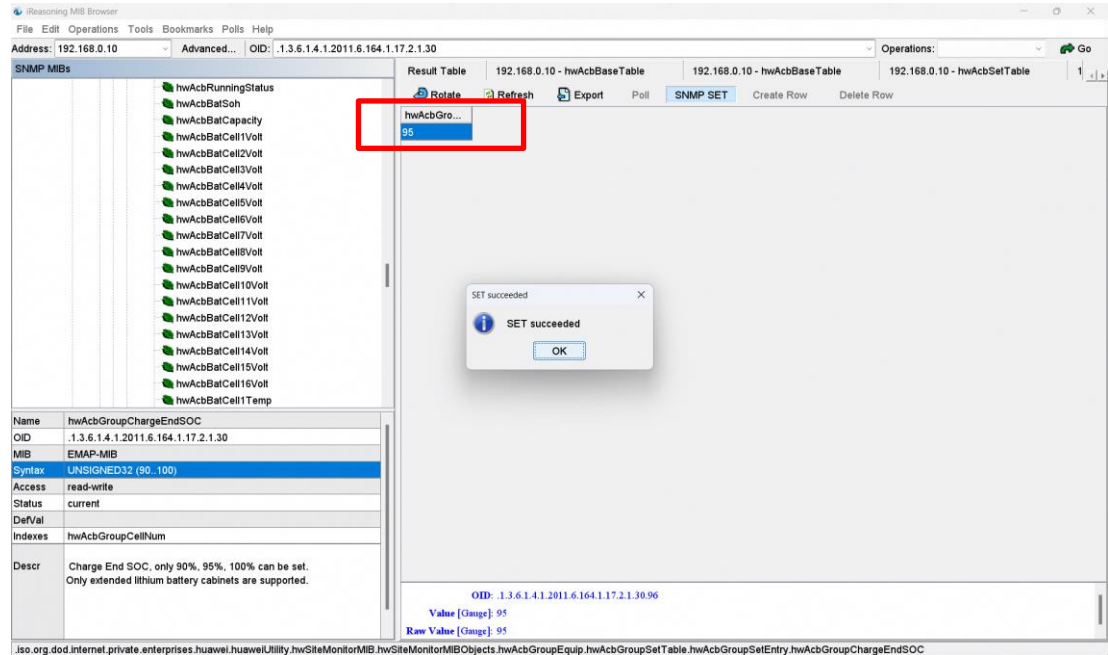
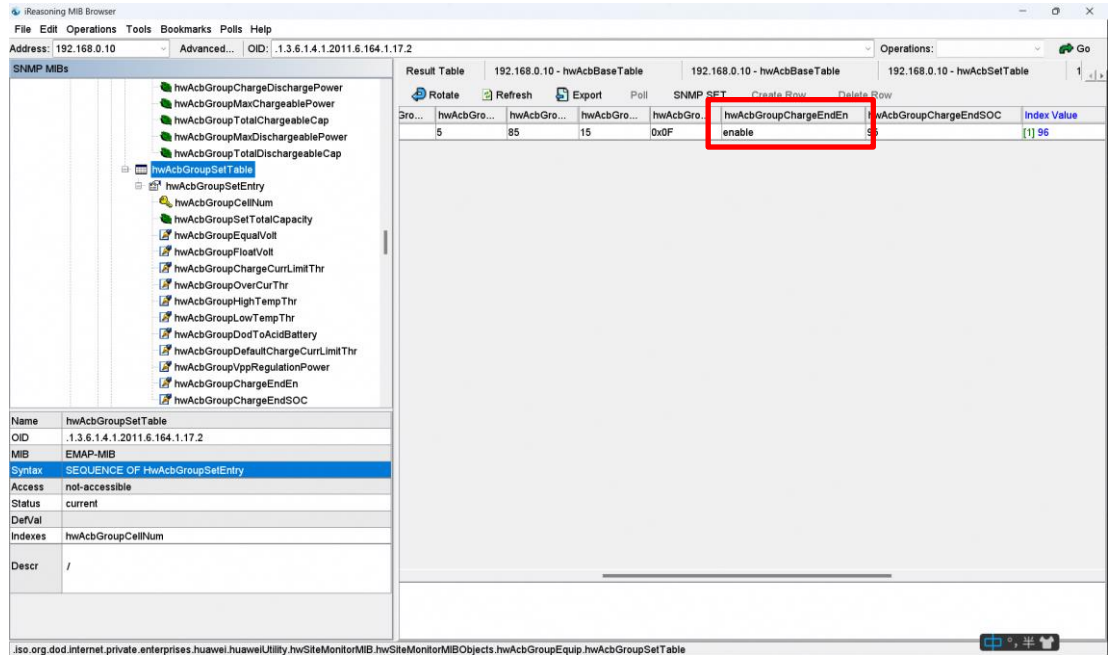
Parameter	Value
Battery Voltage	48.9 V
Battery Current	-23.4 A
Cell Pack Voltage	49.2 V
Cell Pack Current	-23.9 A
SOC	90 %
Cell 1 Temperature	30 °C
Cell 2 Temperature	30 °C
Cell 3 Temperature	30 °C
Cell 4 Temperature	28 °C
Cell 5 Temperature	28 °C
Cell 6 Temperature	28 °C
Cell 7 Temperature	28 °C
Cell 8 Temperature	28 °C
Cell 9 Temperature	28 °C
Cell 10 Temperature	28 °C
Cell 11 Temperature	28 °C
Cell 12 Temperature	30 °C
Cell 13 Temperature	30 °C
Cell 14 Temperature	30 °C
Cell 15 Temperature	30 °C

1. The battery has been discharged to 90% SOC.

New Software Feature Test



■SOC 95% Charging Stop Function Test



1. Changed the battery SOC charge limit function 'hwAcbGroupChargeEndEn' to Enable.
2. Limited the charging SOC to 95% via SNMP MIB Browser.

New Software Feature Test



■SOC 95% Charging Stop Function Test

Running Information		Running Parameter	Running Control
Basic Information			
Battery Status	Float Charge		
Battery Voltage	54.0 V		
Battery Current	33.1 A		
Average Temperature	30 °C		
SOC	95		
Total Discharge Times	3		
Total Discharge Capacity	1.868 kWh		
Total Capacity	200.0 Ah		
BMUs (Normal Communication)	2		
Enable iLock	Disabled		
Charge/Discharge Power	1.80 kW		

1. It was verified that the system switched to Float Charge mode when the SOC reached 95%

New Software Feature Test

■SOC 95% Charging Stop Function Test



The screenshot shows the Enspire power system monitoring interface. The left sidebar lists various components: Site Unit, Digital Power, Lithium Battery Group, GPS, Extend LI Battery Cabinet1, UIM, Lithium Battery1 through Lithium Battery8, and Collection Equipment. The main panel displays 'Basic Information' for the Lithium Battery Group. The SOC is highlighted with a red box at 95%.

Parameter	Value
Battery Voltage	54.0 V
Battery Current	15.2 A
Cell Pack Voltage	50.6 V
Cell Pack Current	16.3 A
SOC	95 %
Cell 1 Temperature	31 °C
Cell 2 Temperature	31 °C
Cell 3 Temperature	31 °C
Cell 4 Temperature	29 °C
Cell 5 Temperature	29 °C
Cell 6 Temperature	29 °C
Cell 7 Temperature	29 °C
Cell 8 Temperature	28 °C
Cell 9 Temperature	28 °C
Cell 10 Temperature	28 °C
Cell 11 Temperature	28 °C
Cell 12 Temperature	31 °C
Cell 13 Temperature	31 °C
Cell 14 Temperature	31 °C
Cell 15 Temperature	31 °C

The screenshot shows the Enspire power system monitoring interface. The left sidebar lists various components: Site Unit, Digital Power, Lithium Battery Group, GPS, Extend LI Battery Cabinet1, UIM, Lithium Battery1 through Lithium Battery8, and Collection Equipment. The main panel displays 'Basic Information' for the Lithium Battery Group. The SOC is highlighted with a red box at 95%.

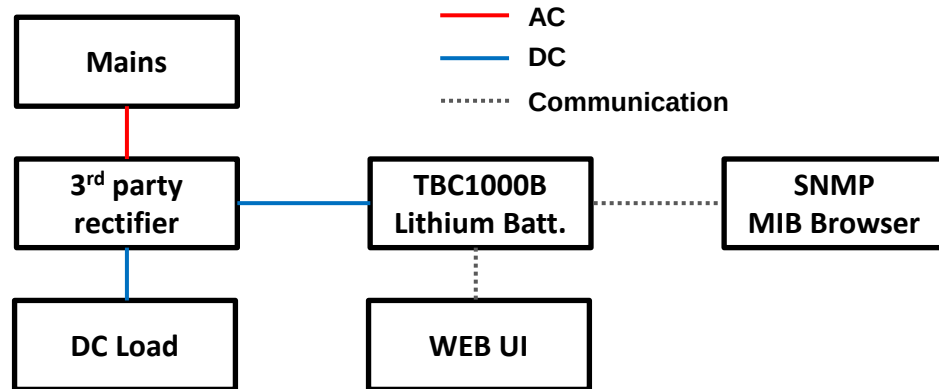
Parameter	Value
Battery Voltage	54.0 V
Battery Current	14.8 A
Cell Pack Voltage	50.6 V
Cell Pack Current	16.1 A
SOC	95 %
Cell 1 Temperature	32 °C
Cell 2 Temperature	32 °C
Cell 3 Temperature	32 °C
Cell 4 Temperature	30 °C
Cell 5 Temperature	30 °C
Cell 6 Temperature	30 °C
Cell 7 Temperature	30 °C
Cell 8 Temperature	29 °C
Cell 9 Temperature	29 °C
Cell 10 Temperature	29 °C
Cell 11 Temperature	29 °C
Cell 12 Temperature	32 °C
Cell 13 Temperature	32 °C
Cell 14 Temperature	32 °C
Cell 15 Temperature	32 °C

1. It was verified that the system switched to Float Charge mode when the SOC reached 95%

New Software Feature Test

■ Battery Remote Disconnect Function Test

□ Test Block diagram



□ Test Method

1. Changed the Control Mode to 'Manual' via the Web UI.
2. Set the battery module to 'OFF' via SNMP MIB Browser.
3. Verify the battery remote disconnect status.
4. Switch the battery to 'Remote ON' and verify normal operation.

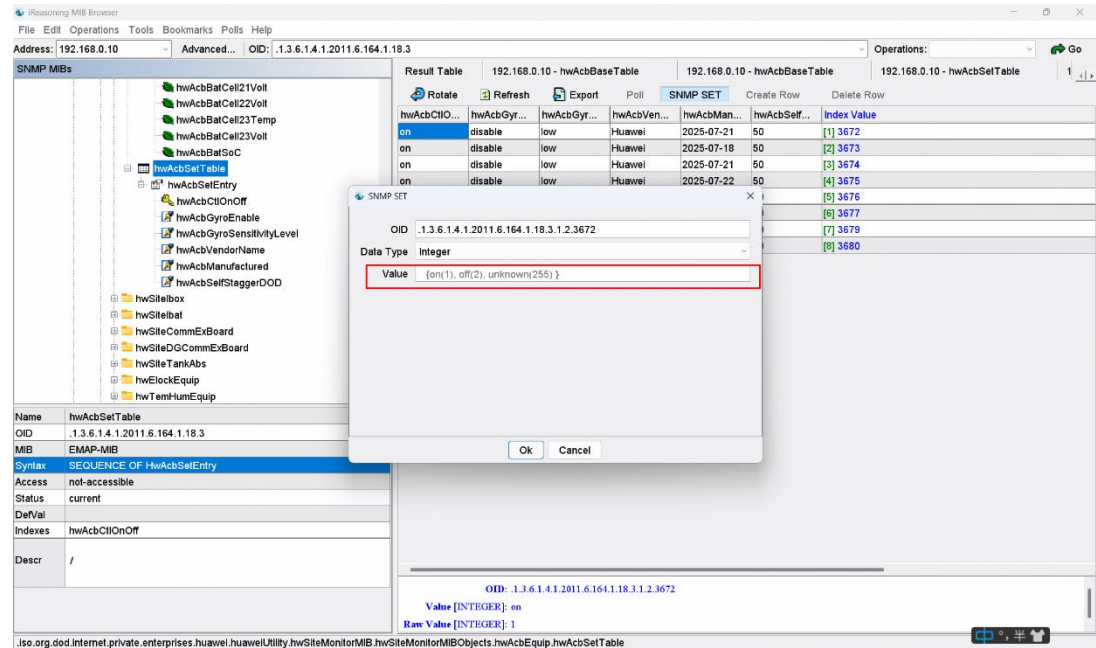
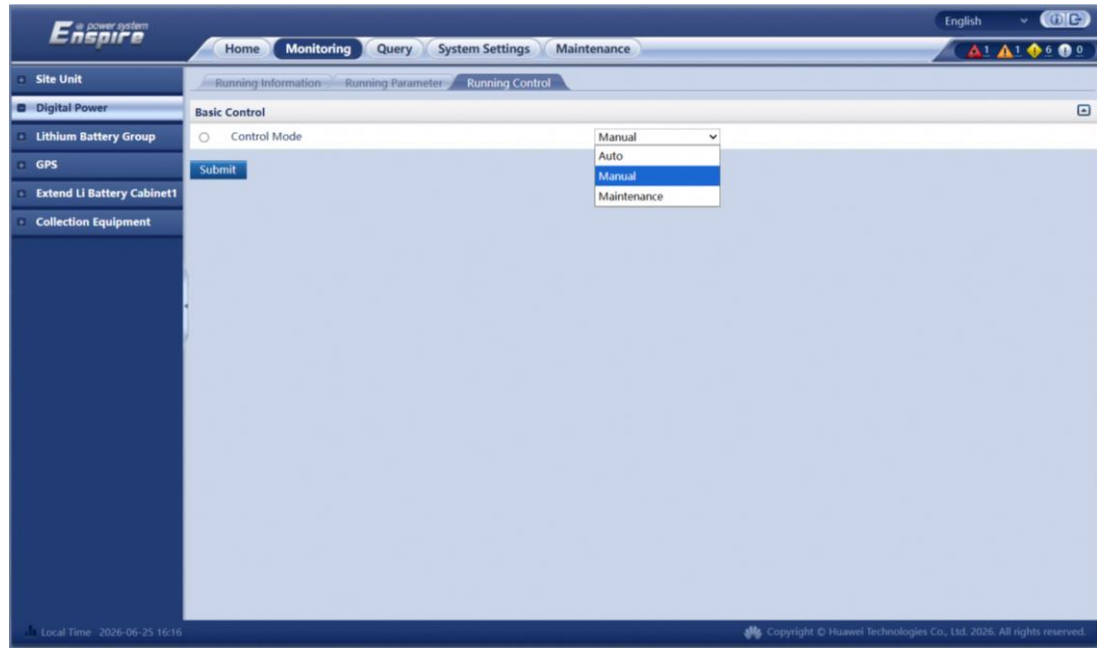
□ Test result

1. "Remote Battery OFF and ON functions are operating normally.
2. **Partial Pass**
3. **The battery does not completely power off but enters an offline mode (sleep mode), and the front LED continues to light up.**

New Software Feature Test



Battery Remote Disconnect Function Test



1. Changed the Control Mode to 'Manual' via the Web UI.
2. Set the battery module to 'OFF' via SNMP MIB Browser.

New Software Feature Test



Battery Remote Disconnect Function Test

4.1.18.3

Result Table 192.168.0.10 - hwAcbBaseTable 192.168.0.10 - hwAcbBaseTable 192.168.0.10 -

Rotate Refresh Export Poll SNMP SET Create Row Delete Row

hwAcbIO...	hwAcbGyr...	hwAcbGyr...	hwAcbVen...	hwAcbMan...	hwAcbSelf...	Index Value
off	disable	low	Huawei	2025-07-21	50	[1] 3672
on	disable	low	Huawei	2025-07-18	50	[2] 3673
on	disable	low	Huawei	2025-07-21	50	[3] 3674
on	disable	low	Huawei	2025-07-22	50	[4] 3675
on	disable	low	Huawei	2025-07-21	50	[5] 3676
on	disable	low	Huawei	2025-07-21	50	[6] 3677
on	disable	low	Huawei	2025-07-21	50	[7] 3679
on	disable	low	Huawei	2025-07-21	50	[8] 3680

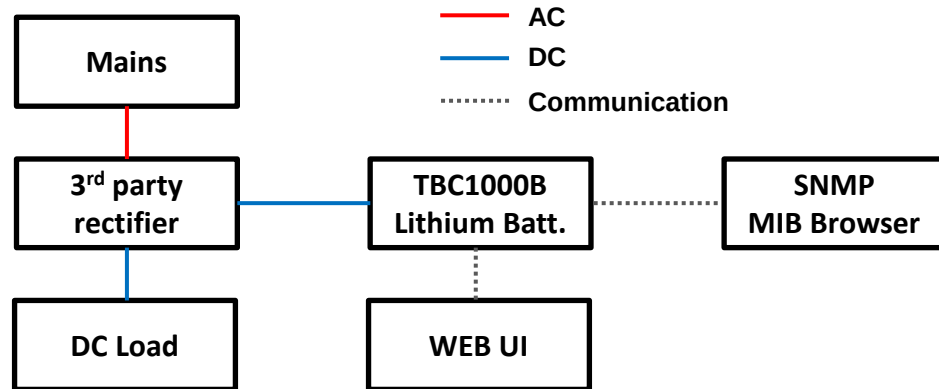


1. After changing the battery to OFF, the module alarm warning light activated and entered Cut-off Mode.
2. The voltage and current values changed to 0, confirming the Cut-off Mode (Web UI screenshots to be added).

New Software Feature Test

■ Checking Battery SOC/SOH/SN via SNMP

□ Test Block diagram



□ Test Method

1. Checking Battery SOC, SOH, and SN using the SNMP MIB Browser.

□ Test result

1. **PASS**

New Software Feature Test



■ Checking Battery SOC/SOH/SN via SNMP

iReasoning MIB Browser

File Edit Operations Tools Bookmarks Polls Help

Address: 192.168.0.10 Advanced... OID: .1.3.6.1.4.1.2011.6.164.1.18.2 Operations: Go

SNMP MIBs

- hwAcbBatCell20Temp
- hwAcbBatCell21Temp
- hwAcbBatCell22Temp
- hwAcbBatCell17Volt
- hwAcbBatCell18Volt
- hwAcbBatCell19Volt
- hwAcbBatCell20Volt
- hwAcbBatCell21Volt
- hwAcbBatCell22Volt
- hwAcbBatCell23Temp
- hwAcbBatCell23Volt
- hwAcbBatSoC
- hwAcbSetTable
 - hwAcbSetEntry
 - hwAcbClOnOff
 - hwAcbGyroEnable
 - hwAcbGyroSensitivityLevel
 - hwAcbVendorName
 - hwAcbManufactured

Result Table

	192.168.0.10 - hwAcbBaseTable	192.168.0.10 - hwAcbBaseTable	192.168.0.10 - hwAcbSetTable	192.168.0.10 - hwAcbSa
...	hwAcbBat...	hwAcbBat...	hwAcbBatSoC	Index Value
17	2147483647	2147483647	100	1 3672
17	2147483647	2147483647	100	2 3673
17	2147483647	2147483647	100	3 3674
17	2147483647	2147483647	100	4 3675
17	2147483647	2147483647	100	5 3676
17	2147483647	2147483647	100	6 3677
17	2147483647	2147483647	100	7 3679
17	2147483647	2147483647	2147483647	8 3680

Rotate Refresh Export Poll SNMP SET Create Row Delete Row

Name hwAcbSampTable

OID .1.3.6.1.4.1.2011.6.164.1.18.2

MIB EMAP-MIB

Syntax SEQUENCE OF HwAcbSampEntry

Access not-accessible

Status current

DefVal

Indexes hwAcbBatVolt

Descr /

OID: .1.3.6.1.4.1.2011.6.164.1.18.2.1.46.3679

Value [UNSIGNED32]: 2147483647

Raw Value [UNSIGNED32]: 2147483647

.iso.org.dod.internet.private.enterprises.huawei.huaweiUtility.hwSiteMonitorMIB.hwSiteMonitorMIBObjects.hwAcbEquip.hwAcbSampTable

1. Checking battery SOC via SNMP MIB Browser.

New Software Feature Test



■ Checking Battery SOC/SOH/SN via SNMP

iReasoning MIB Browser

File Edit Operations Tools Bookmarks Polls Help

Address: 192.168.0.10 Advanced... OID: .1.3.6.1.4.1.2011.6.164.1.18.2 Operations: Go

SNMP MIBs

- hwAcbGroupTotalDischargeableCap
- hwAcbGroupSetTable
- hwAcbGroupSetEntry
- hwAcbEquip
 - hwAcbBaseTable
 - hwAcbBaseEntry
 - hwAcbSampTable
 - hwAcbSampEntry
 - hwAcbBatVolt
 - hwAcbBatCurr
 - hwAcbRunningStatus
 - hwAcbBatSoh
 - hwAcbBatCapacity
 - hwAcbBatCell1Volt
 - hwAcbBatCell2Volt
 - hwAcbBatCell3Volt
 - hwAcbBatCell4Volt
 - hwAcbBatCell5Volt
 - hwAcbBatCell6Volt

Result Table

192.168.0.10 - hwAcbBaseTable		192.168.0.10 - hwAcbBaseTable		192.168.0.10 - hwAcbSetTable		192.168.0.10 - hwAcbSa	
Rotate	Refresh	Export	Poll	SNMP SET	Create Row	Delete Row	
hwAcbBat...	hwAcbBat...	hwAcbRun...	hwAcbBatSoh	hwAcbBat...	hwAcbBat...	hwAcbBat...	hwAcbBat...
542	0	charge	100	1000	339	342	341
541	0	charge	100	1000	343	343	345
542	0	charge	100	1000	337	339	348
542	0	charge	100	1000	336	342	342
542	0	charge	100	1000	345	339	343
542	0	charge	100	1000	347	336	351
542	0	charge	100	1000	350	338	338
2147483647	2147483647	254	2147483647	1000	2147483647	2147483647	2147483647

Name hwAcbSampTable

OID .1.3.6.1.4.1.2011.6.164.1.18.2

MIB EMAP-MIB

Syntax SEQUENCE OF HwAcbSampEntry

Access not-accessible

Status current

DefVal

Indexes hwAcbBatVolt

Descr /

OID: .1.3.6.1.4.1.2011.6.164.1.18.2.1.1.3672

Value [UNSIGNED32]: 542

Raw Value [UNSIGNED32]: 542

.iso.org.dod.internet.private.enterprises.huawei.huaweiUtility.hwSiteMonitorMIB.hwSiteMonitorMIBObjects.hwAcbEquip.hwAcbSampTable

1. Checking battery SOH via SNMP MIB Browser.

New Software Feature Test



■ Checking Battery SOC/SOH/SN via SNMP

iReasoning MIB Browser

File Edit Operations Tools Bookmarks Polls Help

Address: 192.168.0.10 Advanced... OID: .1.3.6.1.4.1.2011.6.164.1.18.1 Operations: Go

SNMP MIBs

- hwAcbGroupTotalDischargeableCap
- hwAcbGroupSetTable
- hwAcbGroupSetEntry
- hwAcbEquip
 - hwAcbBaseTable
 - hwAcbBaseEntry
 - hwAcbSampTable
 - hwAcbSampEntry
 - hwAcbBatVolt
 - hwAcbBatCurr
 - hwAcbRunningStatus
 - hwAcbBatSoh
 - hwAcbBatCapacity
 - hwAcbBatCell1Volt
 - hwAcbBatCell2Volt
 - hwAcbBatCell3Volt
 - hwAcbBatCell4Volt
 - hwAcbBatCell5Volt
 - hwAcbBatCell6Volt

Result Table

hwAcbIndex	hwAcbEqu...	hwAcbEqu...	hw...	hwAcbSoft...	hwAcbFat...	hwAcbHar...	hwAcbModel	hwAcbBarCode	Index Value
3672	31008	Li Battery8	8	V136	Extend Li Bat...	B	ESM-48100C1	UB2572513900	[1] 3672
3673	31001	Li Battery1	1	V136	Extend Li Bat...	B	ESM-48100C1	UB2572511597	[2] 3673
3674	31007	Li Battery7	7	V136	Extend Li Bat...	B	ESM-48100C1	UB2572514202	[3] 3674
3675	31002	Li Battery2	2	V136	Extend Li Bat...	B	ESM-48100C1	UB2572514913	[4] 3675
3676	31003	Li Battery3	3	V136	Extend Li Bat...	B	ESM-48100C1	UB2572513902	[5] 3676
3677	31006	Li Battery6	6	V136	Extend Li Bat...	B	ESM-48100C1	UB2572514199	[6] 3677
3679	31004	Li Battery4	4	V136	Extend Li Bat...	B	ESM-48100C1	UB2572513922	[7] 3679
3680	31005	Li Battery5	5	V133	Extend Li Bat...	B	ESM-48100C1	UB2572514201	[8] 3680

Name hwAcbBaseTable

OID .1.3.6.1.4.1.2011.6.164.1.18.1

MIB EMAP-MIB

Syntax SEQUENCE OF HwAcbBaseEntry

Access not-accessible

Status current

DefVal

Indexes hwAcbIndex

Descr /

iso.org.dod.internet.private.enterprises.huawei.huaweiUtility.hwSiteMonitorMIB.hwSiteMonitorMIBObjects.hwAcbEquip.hwAcbBaseTable

1. Checking battery Serial Number via SNMP MIB Browser.

THANK YOU

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